MAHIMA YUSUF TARANA

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RESEARCH INTERESTS

AI-Driven Flood Risk Modelling, Coastal Hazard Modelling, GeoAI & Machine Learning, Climate Risk & Damage Assessment

ACADEMIC QUALIFICATION

Master of Science in *Disaster Science & Climate Resilience*, University of Dhaka, 2023 - CGPA: **3.76**/4.00
Thesis – Water Quality Prediction of Buriganga River using Remote Sensing Techniques.

Bachelor of Science in *Disaster Science & Climate Resilience,* University of Dhaka, 2022 - CGPA: **3.47**/4.00
Research Project – Assessment of Flood-Induced Health Risks in Louhajang Upazila, Munshiganj, Bangladesh.

TECHNICAL SKILLS

GIS & RS: ArcGIS, QGIS, Google Earth Engine, SNAP, ERDAS IMAGINE.

LiDAR / 3D: DJI Terra, CloudCompare, DEM generation, RTK data integration.

Programming: Python (TensorFlow/Keras, Scikit-learn, Rasterio, GeoPandas, NumPy, Pandas, Matplotlib) & R.

AI / ML & XAI: Random Forest, XGBoost, SVR, GPR, VAE; Integrated Gradients, SHAP/ALE.

RESEARCH & PUBLICATIONS

Published Papers

- Ferdausi, M., **Yusuf Tarana, M.,** Sajadul Alam, M., Fahim Murshed, M., Asif Ehsan, S. M., Lutfe Elahi, M. M., Ahmed, S., Farhadur Rahman, M., Maria Khan, H., Abdul Ahad Badhon, M., Kausar Hossein, M., & Jakariya, M. (2026). Strengthening resilience in ultra-poverty: using multilayered indicators approach in climate-induced vulnerability. International Journal of Sustainable Development & World Ecology, 33(1), 68–80. https://doi.org/10.1080/13504509.2025.2590759
- Tasnim, S. S., **Tarana, M. Y**., Moniruzzaman, D. M., Afrin, S., & Shreya, S. S. (2026). Evaluating Heavy Metal and Physicochemical Pollution in the Karnaphuli River Estuary through Remote Sensing and Machine Learning Approaches. Journal of Earth Observation and Geospatial Applications, 2026(2), 44–63. https://doi.org/10.65372/QGARVS44
- Tasnim, S. S., Hoque, R. M. S., **Tarana, M. Y**., & Rahman, M. M. (2023). Agricultural damage assessment for paddy fields from land use land cover and flood extent maps of five unions of Sunamganj using remote sensing–based analysis. Journal of Bangladesh Institute of Planners, 1–18. https://doi.org/10.3329/JBIP.V16I1.77038
- **Mahima Yusuf Tarana**, Md. Zillur Rahman, Water Quality Prediction of Buriganga River Using Remote Sensing Techniques, in Proceedings of OHOW 2023 – The 2nd International Symposium on One Health, One World, Dhaka University, Dhaka, 6 December–8 December 2023, MDPI: Basel, Switzerland. https://sciforum.net/paper/17452

Selected Manuscripts & Ongoing Research

- Co-author, "Seven-Day Tide Forecasting for Early Warning of Embankment Overtopping in Coastal Bangladesh Using Deep Learning Method." [Submitted]
- Co-author, "Spatiotemporal Flood Inundation Forecasting using a Variational Autoencoder with Explainable AI

in Coastal Bangladesh". [In progress]

- Co-author (2nd), "Airborne LiDAR Reveals Large-Scale Geometric Variability of Flood-Protection Embankments and Its Association with Vegetation Cover in South-eastern Bangladesh". [In progress]
- Author, "Beyond river stage: a multi-source breach prediction framework for deltaic flood embankments combining structural, hydraulic, and anthropogenic drivers". [In progress]
- Author, "Uncertainty-Aware Water Quality Estimation in the Buriganga River using Gaussian Process Regression and Satellite Reflectance." [Upgradation of MS Thesis for Journal Publication - In progress]

RESEARCH EXPERIENCE

Research Associate (North South University, Aug 2023 – Present)
Project: SURF-IT (Funded by FCDO & IDRC under CLARE Programme)

- Developed multi-temporal Sentinel-1/2 flood inundation model and VAE-based spatiotemporal flood forecasting with Integrated Gradients explainable AI.
- Mapped embankment vulnerability by integrating LiDAR-derived DEMs, RTK surveys, soil-strength results and machine-learning classification across coastal Bangladesh.
- Executed WRF-based high-resolution weather downscaling and processed ECMWF meteorological and reanalysis data to support early-warning model development.
- Integrated household survey and spatial datasets to construct socio-economic vulnerability indices for climate-exposed coastal communities.
- Georeferenced historical bathymetry, generated hydro-morphological contour surfaces and produced Planet-based LULC maps for coastal risk analysis.
- Supervised a 5-member team processing 400+ LiDAR point clouds to generate Digital Elevation Model (DEM)
- Facilitated capacity-building workshops with national agencies (BMD, DDM, BWDB, FFWC).
- Represented project outcomes at CLARE GEI Workshop, Kigali, Rwanda (Sept 2024).

Research Assistant (North South University, July 2022 – July 2023)

- Modelled water-quality parameters from georeferenced field samples and Sentinel-2A reflectance; compared the predictability of regression models (MLR, Ridge, Lasso, SVR and PCR) using cross-validation.
- Conducted Sentinel-based flood/agricultural-damage mapping and Sentinel-5P time-series air-pollution analysis.

Research Assistant (University of Dhaka, Apr – Nov 2021)

- Mapped flood inundation and assessed spatial overlap with health outcomes to identify high-risk areas in Munshiganj, Bangladesh.

RESEARCH COLLABORATIONS

- Methodological Advisor — *"A Comprehensive and Explainable GeoAI Framework for Flood Susceptibility Mapping and Spatiotemporal Flood Prediction with Reproducibility in Africa"* (SURF-IT area case study), led by Vivian Asogwa (Sept 2026 – Feb 2027)
  - Advising on the GeoAI methodological framework, including flood-conditioning feature selection (correlation/VIF/RFE), RF and XGBoost susceptibility modeling, and ConvLSTM-based spatiotemporal flood prediction design.

SELECTED TRAINING & CERTIFICATIONS

- Numerical Model & Artificial Intelligence for Ocean Science (NOAMI, 2023)
- Socio-economic & ML-based Environmental Hazard Assessment (NSU, 2023)
- AI-based Vulnerability Assessment Tool & Development of Spatial Data using Drone and Satellite Image (NSU, 2025)

ACADEMIC ACHIEVEMENTS & AWARDS

- National Science & Technology Fellowship, Ministry of Science & Technology, Bangladesh (2022)

CONFERENCE & SEMINARS

- Presented thesis outcome at the **OHOW 2023- The 2nd International Symposium on One Health, One World.**
- Presented undergrad research outcome at the **1st Joint Student Seminar between the University of Tokyo and the University of Dhaka**.

LANGUAGE QUALIFICATIONS

- **Bengali:** Native
- **English:** C1 (IELTS Academic Band 7.0)
- **Spanish:** A1

References

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